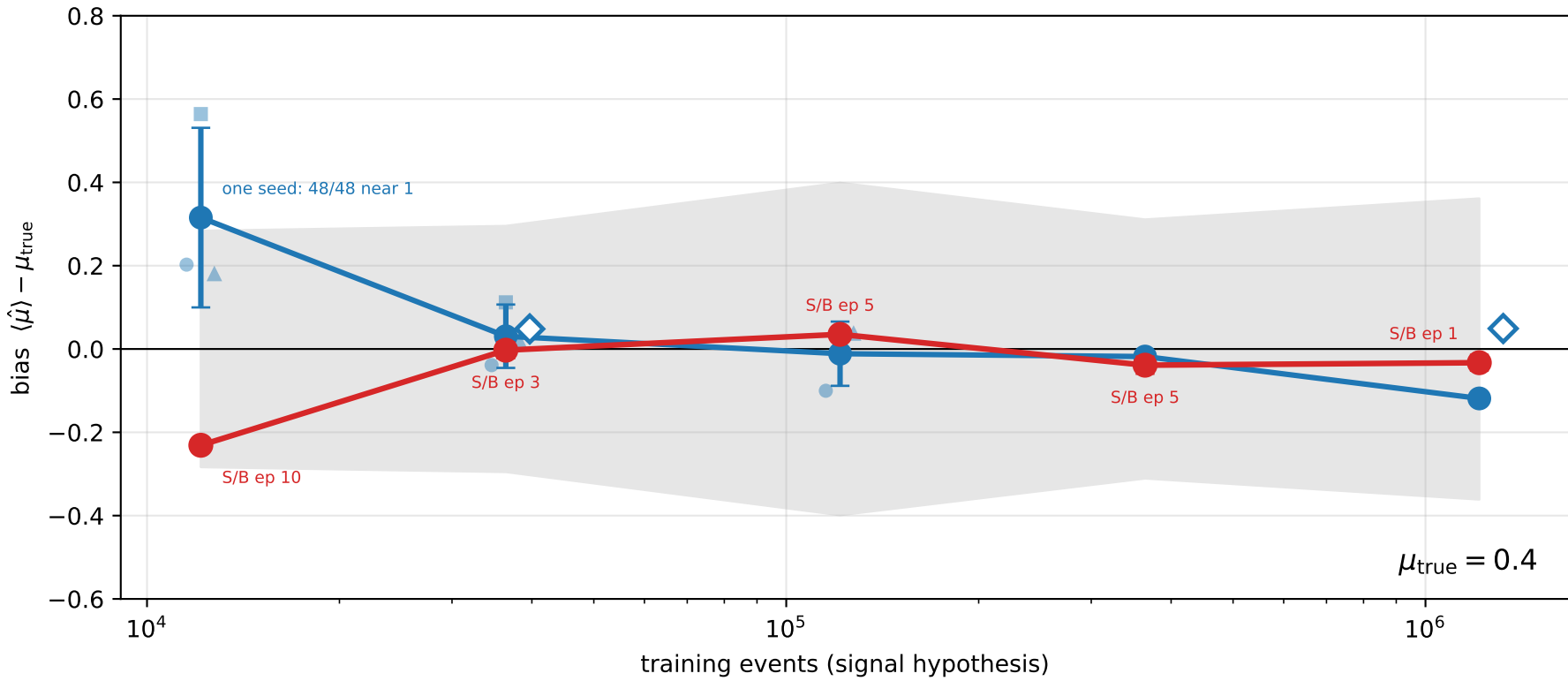
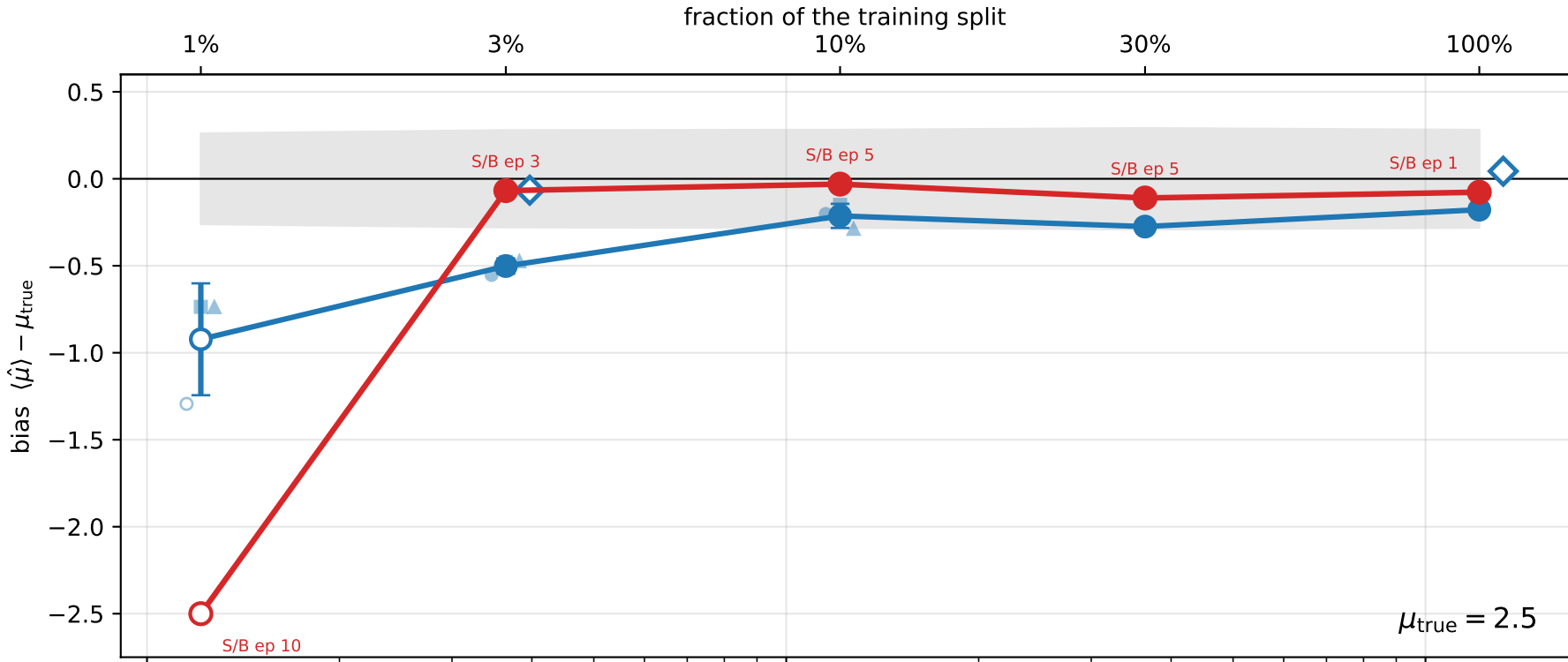


NSBI signal-strength bias vs training statistics: fine-tuned pretrained EveNet vs scratch
48 weighted toys, 300 fb⁻¹



- pretrained, fine-tuned (its own S/B and SBI/B, 20-pass schedule), seed 0: S/B at the smallest-validation-tail checkpoint, SBI/B at epoch 20 (30 %: S/B epoch 5 shown; the rule's pick, epoch 10, gives +0.14 / -0.14)
- scratch (its own S/B and SBI/B, early-stopped best-val): mean over seeds 0/1/2 (small markers) at 1/3/10 %, seed 0 at 30/100 %
- ◇ scratch, a second all-scratch training (annealing test, own S/B and SBI/B, best-val, seed 0):
3 %: same S/B, its SBI/B trained 60k annealed steps instead of 8k; 100 %: a plain retraining
- $\pm \sigma_{\text{stat}}$: half of the mean 1σ width of the scratch fits

error bars: seed spread where 3 seeds exist, otherwise toy s.e. only (excludes run-to-run spread, about 0.07 at 10 %).

Hollow: some 1σ intervals open; points at -2.5: all 48 fits pinned at $\hat{\mu} = 0$.